

SHAHMUN JAFRI

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Mathematics–Computer Science student at UC San Diego (B.S. June 2026) with experience in full-stack, backend, and systems-level development. Built and containerized PostgreSQL/Prisma/Next.js backends with Docker for scalable order-tracking systems, engineered C/C++ memory allocators and custom shells, and applied Python data pipelines for real-world analytics. Skilled in React, TypeScript, Node.js, Golang, and C/C++, with strong foundations in operating systems, algorithms, and optimization. Recognized for combining analytical precision with practical engineering to deliver reliable, efficient, user-focused software.

EDUCATION

UNIVERSITY OF CALIFORNIA, SAN DIEGO

San Diego, CA

Bachelor of Science, Mathematics-Computer Science

June 2026

Relevant Coursework: System Programming and Software Tools, Data Science and Optimization, Machine Learning, Data Science in Practice, Graph Theory and Combinatorics, Advanced **Data Structures and Algorithms, Operating Systems, Computing Theory**

SKILLS

Languages:

Intel x86 Assembly Language, C++, C, Java, Python, HTML, CSS, **Javascript/Typescript**

Operating Systems:

Windows, macOS, Linux

Developer Tools:

Software Engineering Life Cycle, OOP, Valgrind, Git, Github, Vim, VS Code

Software Tools:

React, Next, Prisma, **Node.js**, Docker, PostgreSQL, Seaborn, Matplotlib, Numpy, **Golang**

Soft Skills:

Efficient, Customer service, user support, team collaboration, communication

Certifications:

Stanford Machine Learning Certification

RELEVANT EXPERIENCE

Software Engineering Intern

Sunnyvale, CA

Inprintz

June 2025 – September 2025

- Built a PostgreSQL/Prisma database to digitize order tracking and replace paper workflows.
- Containerized the database with Docker for streamlined deployment and scalability.
- Enabled seamless upload and retrieval of order files (PDFs, images) by building a server-side storage system.
- Implemented Next.js backend APIs for real-time order status updates.
- Improved operational efficiency by digitizing workflows, **reducing manual errors by 100%**, and centralizing order tracking.

Undergraduate Researcher

San Jose, CA

Zaidi Lab, SJSU

July 2023 – July 2024

- Built an Arduino-based plasma device that improved wound treatment efficiency and cut resource use by **15%**.
- Reduced size of program by **30%** allowing less storage and less computations per second.
- Increased Energy efficiency by **25%** by creating an aircooled aluminum radiator for the machine.

Lead Tutor

Sunnyvale, CA

Mathnasium

January 2023 – July 2024

- **Supported 100+ math students**, identifying gaps and tailoring instruction to individual learning needs.
- Awarded Best Tutor for clarity and empathy in guiding students through challenges.
- Coordinated student schedules and progress, ensuring consistency and accuracy in tracking.

PROJECTS

Southern California Wildfire Project | shahmun.com/projects/california-wildfire-project

April 2025

Technology used: Python, Numpy, Seaborn, Pandas

- Processed and merged **~100K records** (CSV + GeoJSON), engineered features, and removed invalids for a clean dataset.
- Analyzed structural and seasonal factors, creating geospatial heatmaps (GeoPandas/Folium) to highlight high-risk regions.
- Documented a reproducible pipeline (cleaning → modeling → reporting) with ethics and bias considerations.

Custom Memory Allocator | shahmun.com/projects/custom-heap-allocator

March 2025

Technology used: C, C++

- Implemented a manual heap allocator in C with custom vmalloc/vmfree, using best-fit placement.
- Managed blocks via metadata headers, supporting splitting and coalescing to reduce fragmentation.
- Added double-free protection, boundary checks, and pointer validation for robustness.

Custom Shell Project | shahmun.com/projects/custom-shell

February 2025

Technology used: C, C++

- Built a mini Unix shell in C/C++ supporting built-in commands, external program execution, and process management.
- Strengthened knowledge of operating system concepts, including process creation, file descriptors, and system calls.
- Added I/O redirection, pipelining, and background execution using process forking and IPC.

VOLUNTEER WORK

- Volunteered at DeltaX to promote and teach Computer Science concepts using scratch to elementary school children.